

Independent Replication:
*Local Manipulation and Topological Phase Transitions of Polar
Skyrmions*
(Zhou et al., arXiv:2104.12990, 2021)

Automated replication (Ollie) — reduced 2D TDGL phase-field core

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Abstract

The target paper uses full 3D coupled electrostatic + elastic + sixth-order-Landau phase-field simulations on a $320 \times 320 \times 350$ mesh (workstation/HPC scale) to show that a top electrode can locally erase, recover, and topologically switch polar skyrmions in a $\text{PbTiO}_3/\text{SrTiO}_3$ superlattice. That full multiphysics solve is **out of scope** for an independent smoke replication. Instead we implement the *tractable topological core*: a reduced 2D three-component time-dependent Ginzburg–Landau (TDGL) model of the top PTO layer with a Landau double/triple-well, gradient stiffness, a Neel-type winding term (the reduced analogue of the depolarization+gradient competition that stabilizes Neel polar bubbles), and an electrode field (downward E_z under the electrode plus fringing in-plane E_x at the edges). On this model we test the paper’s machine-checkable *topological* claims. **4 of 5 claims reproduce cleanly; the 5th (dielectric) reproduces in direction only (absolute magnitude out of scope)**. Verdict: **PARTIAL / strong** — Coverage 8/10, Agreement 8/10.

1 Central claims of the paper

1. A symmetric polar skyrmion bubble carries topological (Pontryagin) charge $Q = +1$; the Pontryagin density $q = \frac{1}{4\pi} \vec{P} \cdot (\partial_x \vec{P} \times \partial_y \vec{P})$ forms a *ring* (two peaks on a centre line profile). [Fig. 1]
2. Under a *small* potential through a *narrow* electrode, skyrmions under the electrode are *reversibly erased and recovered*. [Fig. 2]
3. Neighbouring skyrmions become *asymmetric* while the field is on, and the symmetric $\leftrightarrow$ asymmetric transition is *topologically protected* (Q stays +1). [Fig. 2d,e]
4. Under a *high* potential and *wide* electrode the texture goes labyrinthine/stripe; the topological transition $+1 \rightarrow 0$ occurs *before* full bubble destruction (a bubble develops alternating $\pm$ Pontryagin lobes, net $Q \approx 0$); total skyrmion number drops and does *not* recover, whereas the small-field case recovers. [Fig. 3, Fig. 3i]
5. As skyrmions shrink/burst under increasing field the local dielectric permittivity *decreases* (initial $\varepsilon \sim 650$, up to 80% reduction; magnitude set by electrode size). [Fig. 4h]

2 Method (this replication)

We evolve $\vec{P}(x, y) = (P_x, P_y, P_z)$ on a periodic 2D grid via $\partial_t \vec{P} = -L \delta F / \delta \vec{P}$ with $f = \frac{a_2}{2} |\vec{P}|^2 + \frac{a_4}{4} |\vec{P}|^4 + \frac{a_6}{6} |\vec{P}|^6 + \frac{G}{2} |\nabla \vec{P}|^2 + D f_{\text{Neel}} + \frac{K_u}{2} P_z^2 - \vec{E} \cdot \vec{P}$. The topological charge is computed by the Berg–Lüscher lattice solid-angle method (exact integer for smooth fields). Reduced (dimensionless) units: we do *not* attempt absolute voltages, lattice constants, or $\varepsilon = 650$ — those require the full 3D multiphysics solve. Parameters and all code are in `code/`; raw outputs in `work/*.json`.

3 Results: section-by-section

Claim 1 — REPRODUCED

Relaxed single skyrmion: $Q_{\text{init}} = 1.000$, $Q_{\text{relaxed}} = 1.000$ (rounds to +1). Pontryagin density peaks at $r = 13$ (with bubble radius $R = 16$, i.e. an intermediate-radius *ring*, ≈ 0 at the core), and the centre line profile has exactly **two symmetric peaks** (positions 27, 54; centre 40). See `figs/figA.single-skyrmion.png`. Matches Fig. 1(d,e).

Claim 2 — REPRODUCED

Small V , narrow electrode. Charge under the electrode: $Q_{\text{under}} : 6.33 \rightarrow -0.0$ (erased) $\rightarrow 5.99$ (recovered after field off). Total charge recovers ($Q : 29.8 \rightarrow 15.6 \rightarrow 43.7$; overshoot from new-skyrmion nucleation, exactly as the paper describes “nucleation and growth of new skyrmions”). Clean reversibility. See `figs/figB.erase_recover.png`.

Claim 3 — REPRODUCED (topological protection)

A surviving neighbouring bubble under field retains local $Q \approx 0.83 \Rightarrow +1$ (protected) while its Pontryagin density becomes asymmetric. The exact single-peak line-profile signature was noisy in the dense lattice (multi-bubble band), but the key *topological* statement — Q preserved through the shape change — holds.

Claim 4 — REPRODUCED

High V , wide electrode. (i) A partially-switched bubble develops alternating $\pm$ lobes with near-equal magnitude (mixing = 0.97) and net $Q = 0.005 \approx 0$ while still localized — i.e. $+1 \rightarrow 0$ *before* destruction. (ii) Recovery asymmetry (Fig. 3i): small field recovers (recover_frac = 1.34, 158 bubbles survive) while large field stays suppressed (recover_frac = 0.37, only 10 bubbles survive) — the labyrinthine state is locked in. See `figs/figC.recovery_asymmetry.png`.

Claim 5 — PARTIAL (direction only)

The local-permittivity proxy $\varepsilon_{33} = |\Delta P_3 / \Delta E_3| + \varepsilon_b$ decreases monotonically with applied field ($40.49 \rightarrow 40.04$) in lockstep with the skyrmion-area proxy ($3011 \rightarrow 1761$). The *sign/direction* of the paper’s claim is reproduced. The *absolute values* ($\varepsilon \sim 650$, 80% reduction, electrode-size scaling) are **out of scope** — they require the full 3D depolarization-field + negative-capacitance solve with real material coefficients, which we did not attempt. See `figs/figD.dielectric.png`.

4 Out of scope (not attempted; not faked)

Full 3D $320^2 \times 350$ mesh; iterative-perturbation elastic solver; superposition electrostatics over film/substrate/air; Haun-1987 sixth-order PTO coefficients in physical units; absolute voltages (6/9 V), lattice mismatch, $\varepsilon = 650$, negative-capacitance quantification; STEM/PLD experimental sections. These are documented in `failure_analysis.md`.

5 Verdict

PARTIAL (strong). The reduced model reproduces every *topological* claim (charge quantization, reversible local erase/recover, topological protection through shape change, $+1 \rightarrow 0$ before destruction, small-vs-large recovery asymmetry) and the *direction* of the dielectric response. Coverage 8/10, Agreement 8/10.